

Mark Wu

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EDUCATION

Diamond Bar High School

Diamond Bar, CA

Expected Graduation: June 2026

2024 – Present

- Relevant Coursework: AP Physics C: Mechanics, AP Calculus BC, AP Computer Science A, AP Microeconomics, AP Macroeconomics, AP Precalculus, PLTW
- Currently Taking: AP Chemistry, AP Statistics

ShenWai International School

Shenzhen, China

IB Middle Years Programme

2019 – 2024

- Received IB MYP Bilingual Diploma

PROJECTS

Project Aetherius | ArduPilot, MAVLink, Python, TypeScript

2024 – Present

- Independently designed and built a 2m fixed-wing UAV, covering airframe construction and avionics integration
- Developed a custom multiplatform full-stack ground control system (GCS) built entirely from scratch
- Currently planning on first flight; ongoing tuning and iteration with a small team of collaborators

SprocketStats | TypeScript, Python, C#, SQL, PWA, YOLO26, scikit-learn

2024 – Present

- Built and deployed match prediction pipeline using scikit-learn (Random Forest, K-Means) achieving 90%+ match prediction accuracy and a mean point error of 10 points at competition
- Designed YOLO26-based robot detection pipeline for real-time alliance tracking; currently leading development of multi-target multi-camera tracking (MTMCT)
- Deployed at 3 FRC competitions as sole technical lead managing full development and deployment lifecycle across a 5-person subteam

EXPERIENCE & ACTIVITIES

FRC Team Sprocket 3473

Diamond Bar, CA

CAD Member | Scouting App Technical Lead | Outreach | Driveteam

2024 – 2026

- Design and model robot components using SolidWorks and Fusion 360 as part of the CAD sub-team
- Lead outreach initiatives to engage the local community and promote STEM involvement

Dronescape Club — Engineering Lead

Diamond Bar, CA

Diamond Bar High School

2025 – Present

- Lead engineering direction for the school drone club, overseeing technical projects and mentoring 4–5 members
- Guide members through hands-on UAV design, build, and flight processes

Team Infernope — Captain

ShenZhen, CN

Combat Robotics

2021 – 2024

- Founded and captained a combat robotics team, managing design, build, and competition strategy
- Won the 2024 school combat robotics tournament

FTA Volunteer — SoCal FTC Qualifiers

Diamond Bar, CA

FIRST Tech Challenge

January 2026

- Served as FIRST Technical Advisor (FTA) at two consecutive SoCal FTC Qualifier events hosted at Diamond Bar High School
- Responsible for field hardware, software, and ensuring matches ran smoothly for competing teams

AWARDS & RECOGNITION

JPL Invention Challenge | The Scavengers

2025

- Competed in the JPL Invention Challenge, designing a fluid transport system without pumps; placed one spot from finals

FRC Team Sprocket 3473

2024, 2025, 2026

- FIRST Impact Award recipient at regionals in 2024 and 2025
- Engineering Inspiration Award recipient at regionals and districts in 2024 and 2026

Team Infernope

2024

- Won 2024 school combat robotics tournament, Shenzhen, China